

Complex Analysis — Sequences and Series (Section 4.1)

Definitions Summary

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Complex Sequences

Complex sequence. A function $z : \mathbb{N} \rightarrow \mathbb{C}$, written $\{z_n\}$.

Limit of a sequence (Def. 4.1.1).

$$\lim_{n \rightarrow \infty} z_n = \zeta$$

if for every $\varepsilon > 0$ there exists N_ε such that

$$|z_n - \zeta| < \varepsilon \quad (n > N_\varepsilon).$$

Boundedness & Cauchy

Bounded sequence (Def. 4.1.7). There exist $R > 0$ and N such that

$$|z_n| < R \quad (n > N).$$

Cauchy sequence (Def. 4.1.9). For every $\varepsilon > 0$ there exists N_ε such that

$$|z_n - z_m| < \varepsilon \quad (n, m > N_\varepsilon).$$

Infinite Series

Infinite series (Def. 4.1.11).

$$\sum_{k=1}^{\infty} z_k = z_1 + z_2 + \cdots$$

Convergence (Def. 4.1.12). Let $S_n = \sum_{k=1}^n z_k$. If $\lim_{n \rightarrow \infty} S_n = S$, then

$$\sum_{k=1}^{\infty} z_k = S.$$

Absolute convergence (Def. 4.1.13).

$$\sum_{k=1}^{\infty} |z_k| \quad \text{converges.}$$

Divergence (Def. 4.1.14). A series diverges if it does not converge.

Products of Series

Cauchy product (Def. 4.1.22). Given $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$,

$$\sum_{n=0}^{\infty} c_n, \quad c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Key idea: Complex convergence is governed by the modulus $|\cdot|$, extending the real theory.